

Assignment Questions: Unit V – I/O and Memory Organization

Part A:

1. Define **Input–Output (I/O) organization** in computer systems.
2. What is an **I/O interface**?
3. Define **asynchronous data transfer**.
4. What are the different **modes of data transfer**?
5. What is meant by **interrupt-driven I/O**?
6. Define **priority interrupt**.
7. What is **Direct Memory Access (DMA)**?
8. State the advantages of **DMA** over programmed I/O.
9. Define **memory hierarchy**.
10. What is **main memory**?
11. Differentiate between **primary memory** and **secondary memory**.
12. Define **auxiliary memory** with examples.
13. What is **associative memory**?
14. Define **cache memory**.
15. What is the importance of **cache memory** in computer systems?

Part B:

16. Explain the concept of **Input–Output interface** with a neat diagram.
17. I/O interfaces are always included in digital computers. Why is it so important?
18. Describe Input-Output Processor (IOP).
19. Discuss the working principle of **asynchronous data transfer**.
20. Explain different **modes of data transfer** in computer systems.
21. Describe the concept of **priority interrupts** and their significance.
22. Explain the **Direct Memory Access (DMA)** technique with suitable examples.
23. Discuss the advantages and disadvantages of **DMA**.
24. Explain the **memory hierarchy** in detail.
25. Discuss the structure and functions of **main memory**.
26. Explain the role of **auxiliary memory** in computer organization.
27. Describe the concept and working of **associative memory**.
28. Explain the structure and functions of **cache memory**.
29. Discuss how **cache memory improves system performance**.
30. Differentiate between:
 - a) Synchronous and asynchronous data transfer
 - b) Main memory and auxiliary memory
 - c) Cache memory and associative memory
 - d) Interrupt-driven I/O and DMA
31. Analyze the role of **I/O interfaces** in communication between CPU and peripheral devices.
32. Explain how **priority interrupts** improve system efficiency.